Wednesday Warm Up: Unit 0, Review of 135A

Prof. Jordan C. Hanson

January 14, 2026

1 Review of 135A

1. Suppose a physical therapy patient is asked to shove a medicine ball forward off the edge of a table to help rebuild the strength of their shoulders. The medicine ball weighs 7 kg. (a) If the patient is able to give the ball a speed of 1 m s^{-1}, what is the momentum of the ball? (b) If the patient gives the ball the same momentum by rolling it, and it strikes elastically a ball with a mass of 3.5 kg, what will be the velocity of the second ball?

2. Suppose the medicine ball in the previous problem has a mass of 3.5 kg, and a diameter of 10 cm. The moment of inertia for a solid sphere is $I = \frac{2}{5}mr^2$. (a) What is the angular momentum of the ball if it is spun at 1 rotation per second? (b) What is the rotational kinetic energy?

2 Unit 0: Electrostatics I

1. The SI unit of **charge** is called a *Coulomb* (1 C). A Coulomb is the amount of charge that flows through a wire in one second, given that the *current* is 1 *Amp*. Define current as $I = \Delta Q/\Delta t$, where Q is the charge as a function of time, and ΔQ is the change in the charge. (a) Suppose we are charging a car battery with an average current of $I = \Delta Q/\Delta t = 2$ Amps, for 5 hours. How many Coulombs of charge did we put in the battery? (b) If a battery has 7 Amp hr of charge, for how long can it provide 0.5 Amps of current?

3 Mathematics Review

1. Compute the dot-product, $\vec{a} \cdot \vec{b}$.
   - $\vec{a} = -2\hat{i} + 2\hat{j}$, $\vec{b} = 4\hat{i} + 4\hat{j}$.
   - $\vec{a} = 2\hat{i} - 1\hat{j}$, $\vec{b} = -2\hat{i} - 3\hat{j}$.
2. Compute the cross-product, $\vec{a} \times \vec{b}$.
   - $\vec{a} = -2\hat{i} + 2\hat{j}$, $\vec{b} = 4\hat{i} + 4\hat{j}$.
   - $\vec{a} = 2\hat{i} - 1\hat{j}$, $\vec{b} = -2\hat{i} - 3\hat{j}$.